

Development of Intelligent Zebra Crossings to Mitigate Traffic Accidents

Sujit Kumar

Department of Electrical Engineering
Government Engineering College, Banka
Bihar Engineering University, Science, Technology
and Technical Education Department Bihar
✉ sujitaryan0412@gmail.com

Deeba Ashique

Dept. of Electronics and Communication Engineering
Government Engineering College
Banka, Bihar
✉ deeba0891@gmail.com

Md. Zafar

Department of Electrical Engineering
Government Engineering College
Banka, Bihar
✉ mdzafarsabour35@gmail.com

Alka Raj

Department of Computer Science Engineering
Government Engineering College
Banka, Bihar
✉ manishkumar181605@gmail.com

ABSTRACT

This study focuses on improving road safety by introducing innovative solutions to reduce accidents caused by speeding and reckless driving, especially in high-risk areas like highways, hospitals, and schools. With road traffic accidents causing around 1.3 million deaths and millions of injuries annually, the limitations of traditional traffic control systems are becoming increasingly evident. Poor visibility, bad weather, and driver distractions further reduce the effectiveness of conventional signs and signals. To address this, we propose a “smart zebra crossing” system that uses ultrasonic sensors and LED lights to provide real-time alerts to both drivers and pedestrians, improving visibility and safety, especially in low-light conditions. These systems are connected to a central traffic management network, allowing for real-time speed limit adjustments and sending alerts to connected vehicles. This approach significantly enhances road safety by providing an intelligent, adaptive solution for better traffic management and accident prevention.

KEYWORDS: *Automation, Microcontroller, Sensors, Traffic signal, Road safety.*

INTRODUCTION

Urbanization is growing quickly, with more people expected to live in cities by 2030 [1]. This increase in city populations is putting pressure on transportation systems, which are essential for economies and daily life. As the number of vehicles rises, traffic problems like congestion, accidents, and poor infrastructure become more serious [2]. Road accidents have become a major global concern, causing about 1.3 million deaths each year, especially among pedestrians and cyclists. Speeding is a leading cause, making roads less safe [3]. Traditional traffic systems are struggling, highlighting the need for smarter, adaptive traffic solutions. Ahmad Malek Nada et al. [4] proposed a system to reduce Algeria’s high driver mortality rate,

caused by speeding and outdated road conditions. Their solution includes intelligent traffic signs and a smart speed limiter, tested with the SUMO simulator and an Arduino-based prototype, showing significant potential in lowering fatalities. Mohamed [5] explored integrating IoT and Big Data into traffic control systems for enhanced safety and sustainability, highlighting the impact of self-driving cars and intelligent traffic lights in reducing emissions and congestion. Anurag Patil et al. [6] introduced a system using spike speed breakers and traffic signals to optimize traffic flow and safety, featuring real-time signal adjustments. Sandeep Sharma et al. [7] developed a microcontroller-RFID-based system to reduce vehicle speed in accident-prone areas, incorporating automatic braking and a GSM

penalty system. Taeiba Taher et al. [8] presented smart traffic signals with sonar sensors to prevent collisions, while Tan et al. [9] stressed the environmental benefits of smart city transportation systems. Figueiredo et al. [10] highlighted the role of intelligent transportation systems in reducing congestion and improving safety in smart cities. Zhang et al. [11] emphasized the potential of connected vehicles to reduce accidents. Liu et al. [12] and Wang et al. [13] discussed the importance of advanced traffic systems in lowering emissions and promoting sustainable transportation.

This paper presents a novel “smart zebra crossing” system aimed at improving road safety, particularly in urban areas. The system uses ultrasonic sensors to activate embedded LED lights, enhancing visibility for both drivers and pedestrians, especially in low-light conditions. This dynamic system allows authorities to monitor and adapt to changing road conditions, thereby improving traffic management and reducing accidents. The solution is especially effective in high-risk zones like school areas and hospitals. The paper is structured as follows: Section 1 introduces fault diagnosis approaches, Section 2 explains the methodology, Section 3 discusses results, and Section 4 concludes.

METHODOLOGY

This work explores the development of advanced systems such as smart zebra crossings system. This research aimed to enhance pedestrian safety through the development of a smart zebra crossing system that integrates advanced electronic components. The smart zebra crossing differs from traditional crossings by using red LED lights and ultrasonic sensors positioned on both sides. The ultrasonic sensors detect approaching pedestrians, triggering a sequence of red LED lights to illuminate as the pedestrian crosses, providing real-time visual alerts to drivers. This innovative system was designed to improve visibility during low-light or adverse weather conditions, thus addressing safety concerns that traditional crossings, which rely solely on painted lines, cannot adequately address. To implement the system, we utilized Proteus software for simulation, creating a comprehensive operational flow as illustrated in the flowchart (Fig. 1). The system starts with an initialization phase where all components, including ultrasonic sensors and LED lights, are activated. In

standby mode, the system continuously monitors for pedestrian movement. Upon detection, the lights are activated to alert oncoming vehicles, ensuring higher visibility. Once the pedestrian has crossed and no further movement is detected, the system reverts to standby mode.

The Key findings from the simulation indicated the effectiveness of the smart zebra crossing in real-time scenarios. The “Crossing OFF” mode allows normal traffic flow, while the “Crossing ON” mode enhances pedestrian safety by alerting drivers when pedestrians are present. The sequential illumination of the LED lights significantly increases visibility, thereby reducing the risk of accidents. The results demonstrated that the smart crossing system could be particularly beneficial in high-risk areas like schools and hospitals. The research highlights the feasibility of deploying this system in real-world urban environments, paving the way for future enhancements, including integration with central traffic management systems for optimized traffic flow and safety. This section includes smart zebra crossing, circuit diagram and components used for hardware connection.

Smart Zebra Crossing

The smart zebra crossing enhances the conventional pedestrian crossing by incorporating advanced electronic components. Unlike traditional crossings, which rely solely on painted lines, the smart zebra crossing is equipped with red LED lights and ultrasonic sensors on both sides. Fig. 1 shows the operational process of a smart zebra crossing system, focusing on pedestrian detection and the activation of lights.

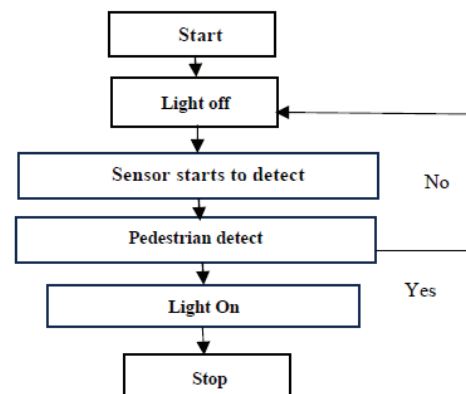


Fig. 1. Flowchart of smart zebra crossing

The system begins with an initialization phase, where all sensors and components are activated and prepared for operation. Initially, the lights remain off, representing no pedestrian activity. The system then enters a standby mode, where ultrasonic or motion sensors continuously monitor the area for pedestrian movement near the zebra crossing. When a pedestrian is detected, the system moves forward, activating the lights embedded within the zebra crossing to alert oncoming vehicles. This increases pedestrian visibility, particularly during low-light conditions or adverse weather. If no pedestrian is detected, the system continues monitoring. Once the pedestrian has safely crossed, and no further movement is detected, the system enters a stop state, and the lights turn off, returning the system to its initial standby mode, ready for the next detection. This operational flow ensures the smart zebra crossing system is efficient, activating the lights only, when necessary, thus enhancing road safety for pedestrians.

Connection Diagram

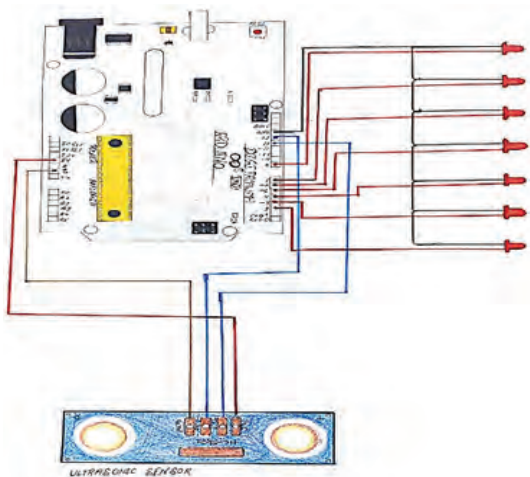


Fig. 2. Circuit Diagram of Smart Zebra Crossing

The ultrasonic sensor is connected to the Arduino UNO through several key pins. The echo pin, which generates a pulse when it receives a reflected signal, is connected to pin 13 on the Arduino. The trigger pin, responsible for initiating the ultrasonic sound wave, is linked to pin 12. For power, the VCC pin of the ultrasonic sensor is connected to the 5V pin on the Arduino, while the GND pin, which serves as a reference point with zero voltage, is connected to the GND pin of the Arduino UNO. This configuration ensures the sensor operates effectively

within the smart zebra crossing system. Fig. 2. Shows the Circuit Diagram of Smart Zebra Crossing.

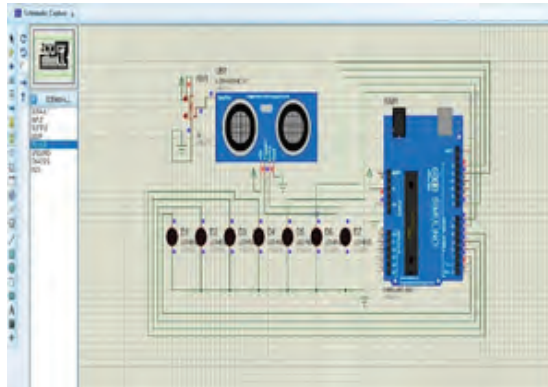
The LED lights are connected to the Arduino UNO to create a sequential illumination system. Specifically, LED 1 is linked to digital pin 2, LED 2 to pin 3, LED 3 to pin 4, LED 4 to pin 5, LED 5 to pin 6, LED 6 to pin 7, and LED 7 to pin 8. This arrangement allows the Arduino to control each LED individually, enabling the smart zebra crossing system to effectively signal pedestrian presence and enhance visibility for drivers. This configuration outlines the sequential connection of multiple LEDs to the respective digital pins of the Arduino Uno for controlled operation in the system.

RESULTS AND DISCUSSION

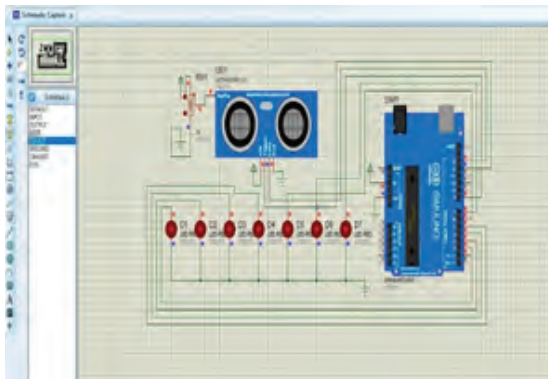
In this work, the proposed smart zebra crossing system was implemented through both software simulation and hardware setup. The system was simulated using Proteus software, focusing on two key operational modes: "Crossing OFF" and "Crossing ON." Fig. 3 (a) and (b) show the simulation diagrams, highlighting the different states of the system.

In the "Crossing OFF" mode, when no pedestrians are detected, the system remains inactive. The LED lights embedded in the crossing stay off, allowing vehicles to pass through without any disruptions. The system is in standby mode, continuously monitoring the area using ultrasonic sensors for pedestrian movement. This ensures smooth traffic flow when there are no pedestrians, avoiding unnecessary delays. When a pedestrian approaches, the ultrasonic sensor detects them, switching the system to the "Crossing ON" mode. The LED lights turn on, signalling both pedestrians and drivers. This improves visibility, especially during low-light conditions, ensuring pedestrian safety. The system alerts drivers to slow down or stop while the pedestrian crosses. After the pedestrian has crossed, the system automatically returns to the "Crossing OFF" mode, resuming normal traffic flow. The simulation results demonstrate the effectiveness of the smart zebra crossing in real-time situations. The system's ability to switch between modes ensures both safety for pedestrians and smooth traffic flow for vehicles. The integration of ultrasonic sensors and LED lights significantly enhances visibility for drivers, reducing the likelihood of accidents.

This system has strong potential to improve road safety, especially in high-risk areas like schools, hospitals, and busy urban zones. The results validate the feasibility of the design for real-world application, offering an adaptive and cost-effective solution to road safety challenges. Fig. 4 shows the hardware connection of the proposed system, highlighting the system's practical implementation and its safety benefits in critical areas.



(a)



(b)

Fig. 3. (a) Smart Zebra Crossing OFF Stage (b) Smart Zebra Crossing ON Stage



Fig. 4. Hardware Connection

CONCLUSION

This work presents a smart zebra crossing system aimed at enhancing road safety through the integration of modern sensing technology. The proposed system employs ultrasonic sensors to detect pedestrian presence and activates embedded LED lights in real time, significantly improving crossing visibility, particularly under low-light conditions or adverse weather. Unlike traditional pedestrian crossings that depend solely on painted markings, this intelligent design dynamically illuminates sequential red LEDs when pedestrian movement is detected, ensuring that approaching drivers receive timely visual cues to slow down and stop. By offering an active rather than passive warning mechanism, the system reduces the risk of pedestrian accidents. Looking forward, further enhancements could include the integration of advanced traffic management technologies, such as AI-driven predictive analytics, to optimize traffic flow and adapt to varying traffic densities. Additionally, combining smart zebra crossings with speed breakers may provide an even more comprehensive and effective solution to improving pedestrian safety in high-risk areas.

REFERENCES

1. Hartig, Terry and Kahn Jr, Peter H.: Living in cities, naturally, Science, vol. 352, no. 6288, pp. 938-- 940, (2024).
2. Gonzalo-Orden, Hernán, Heriberto Pérez-Acebo, Alaitz Linares Unamunzaga, and Marta Rojo Arce. Effects of traffic calming measures in different urban areas. Transportation Research Procedia, vol. 33, pp. 89--90. Elsevier (2018).
3. Abdulmawjoud, Ayman A. and Jamel, Mohammed G., and Al-Taei, Abdulkhalik A.: Traffic flow parameters development modeling at traffic calming measures located on arterial roads. Ain Shams Engineering Journal, vol. 12, no. 01, pp. 437--444. Elsevier (2021).
4. Nada, AHMED MALEK and Rachid, BOUDOUR: Intelligent Brake Controller Based on Intelligent Highway Signs to Avoid Accidents on Algerian Roads. International Journal of Advanced Computer Science and Applications, vol. 14, no. 05. Science and Information (SAI) Organization Limited (2023).
5. Abdelati, Mohamed H.: Smart Traffic Management for Sustainable Development in Cities: Enhancing Safety

- and Efficiency. International Journal of Advanced Engineering and Business Sciences, vol. 05, no. 01. Integrated Thebes Institutes (2024).
6. Patil, Anurag and Kute, Akash and Yadav, Payal and Mhaske, Avinash: SMART TRAFFIC AUTOMATION USING SPIKE SPEED BRAKERS. International Journal of Progressive Research in Engineering Management and Science, vol. 03, pp. 28--33, (2023).
 7. Sharma, Sandeep and Kishore, India Sumit and Mishra, Shrey and Girotra, Dhruv and Talwar, Hardik: Automatic vehicle speed reduction \& accident prevention system using RF technology. Int. Journal of Engg. Applied Sc. and Technology, vol. 05, no. 07, pp. 292--295, (2020).
 8. Taher, Taieba and Ahmed, RU and Haider, MA and Das, Swapnil and Yasmin, MN and Mamun, Nurasdul: Accident prevention smart zone sensing system. 2017 IEEE Region 10 Humanitarian Technology Conference (R10-HTC), pp. 67--70. IEEE (2017).
 9. Trindade, Evelin Priscila and Hinnig, Marcus Phoebe Farias and da Costa, Eduardo Moreira and Marques, Jamile Sabatini and Bastos, Rogério Cid and Yigitcanlar, Tan: Sustainable development of smart cities: A systematic review of the literature. Journal of Open Innovation: Technology, Market, and Complexity, vol. 03, no. 03, pp. 1--14. Elsevier (2017).
 10. Figueiredo, K. A., Júnior, H. A. R., & Oliveira, L. B.: Intelligent transportation systems and smart cities: A systematic literature review: Journal of Cleaner Production, vol. 243, pp. 118630 (2020).
 11. Zhang, Y., Bai, Y., Zeng, L., & Han, J.: Intelligent transportation systems for connected and automated vehicles: A review. Transportation Research Part C: Emerging Technologies, vol. 123, pp. 102998, (2021).
 12. Liu, Y., Ma, X., Wu, Y., & Wang, H.: Traffic flow management in smart cities: An overview. IEEE Transactions on Intelligent Transportation Systems, vol. 21, no. 1, pp. 329--341, (2020).
 13. Wang, S., Liu, J., Wang, M., & Yan, J.: Evaluating the effects of smart transportation on energy consumption and greenhouse gas emissions: A case study of China: Applied Energy, 283, pp. 116196, (2021).